1. **Calculating Factorials:** Demonstrated above.

Fibonacci Series:

```
def fibonacci(n):
    if n <= 1:
        return n
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

print(fibonacci(6))  # Outputs: 8
```

2.
3. **Solving Tower of Hanoi:** A classic problem in computer science that involves moving disks between rods with specific rules.

4. **Tree Traversals:** Recursion is widely used in navigating hierarchical structures like trees and graphs.

Advantages and Disadvantages of Recursion: